

# ALBERTUS BERNAT SENTOSA

**MECHANICAL ENGINEERING GRADUATE | ROBOTICS, AUTOMATION & ADDITIVE MANUFACTURING**


+62-812-9904-3141 | https://ber-sento.github.io/

sentosabernat@gmail.com | Banten, Indonesia


## PROFILE

Mechanical Engineering graduate from Atma Jaya Catholic University (GPA 3.88/4.00, *cum laude*; Best Graduate Ring Award). Experienced in independent software development for five-axis 3D printing, robotic-arm prototyping, and experimental research on hydrogen fuel cells. Interested in robotics, manipulation and control, human–robot interaction, and intelligent automation.

## EDUCATION

### Atma Jaya Catholic University of Indonesia

Bachelor of Mechanical Engineering (S.T.) - November 2021 - July 2026

- GPA: 3.88 / 4.00 - *cum laude*
- Best Graduate Ring Award – top graduate in the Mechanical Engineering program
- Thesis defense grade: A
- Relevant coursework: Robotics, Automatic Control, Mechatronics, Numerical Analysis, Linear Algebra, Finite Element Methods, and PLC.

## RESEARCH & PUBLICATION

### Undergraduate Thesis – DEVELOPMENT OF 3D MODEL SLICER FOR A 5-AXIS 3D PRINTER | 2025–2026

- Independently developed the complete slicer software, from algorithm research and segment-based non-planar slicing to G-code transformation and GUI implementation.
- Integrated PrusaSlicer and developed matrix-based coordinate transformations for five-axis printing.
- Evaluated physical prints, achieving average dimensional deviation of 0.29 mm (0.95%) and angular deviation of 0.5° in reported measurements.

### Published Research – Hydrogen Fuel Cell: Experimental Study of the Generated Voltage | 2025

- Arbiyani, F., Sentosa, A. B., & Limmuel, E. B. JST (Jurnal Sains dan Teknologi), 14(3), 630–642.
- Conducted background research on hydrogen fuel-cell principles and electrochemical reactions.
- Prepared and typed the initial research manuscript; final revisions were made by the supervising professors.
- Independently sourced and purchased materials for the experimental research.
- DOI: https://doi.org/10.23887/jst-undiksha.v14i3.102174

## INDUSTRIAL EXPERIENCE

### Robotics and Automation Intern – PT Cyber Teknologi Nusantara

July – September 2025

- Worked on an EV charging automation prototype using a Kinova Gen3 6-DoF robotic arm and Kinova-based control software.
- Developed an application for the prototype, designed and 3D-printed the charging station, vehicle charging-port model, and charging gun for simulation.
- Supervisor evaluation: Baik Sekali (Very Good) for fieldwork and internship report.

## AWARDS & OTHER ACHIEVEMENTS

- Best Graduate Ring Award, Atma Jaya Catholic University, 2026.
- 1st Place, Best Prototype, and Best Poster, UTM TechVenture Challenge, Universiti Teknologi Malaysia, 2024.
- 2nd Place, Product Innovation Competition, Atma Jaya Business and Innovation Expo, 2024.
- Invited Speaker, 3D Printing Career Day, SMA Kristen Penabur Harapan Indah Bekasi, 2025.
- Best Rank Admission Scholarship, Atma Jaya Catholic University, 2022 – 100% tuition-fee reduction.

## TECHNICAL SKILLS

- Programming: Python, C++, Scilab, GNU Octave.
- Robotics & Automation: Kinova Gen3, robotic kinematics (forward/inverse kinematics, Denavit–Hartenberg), Siemens LOGO! Soft Comfort (PLC ladder programming), Arduino.
- Design & Manufacturing: SolidWorks (~3 years), Autodesk Fusion, Tinkercad, PrusaSlicer, FDM and five-axis 3D printing.
- Simulation: ANSYS (introductory experience).

## LANGUAGE

Indonesian (Native) | English (TOEFL iBT - 100/120 - CEFR C1) | Japanese (JLPT N5 passed)